

CY 510-01 OrbitSec: Functional and Security Requirements

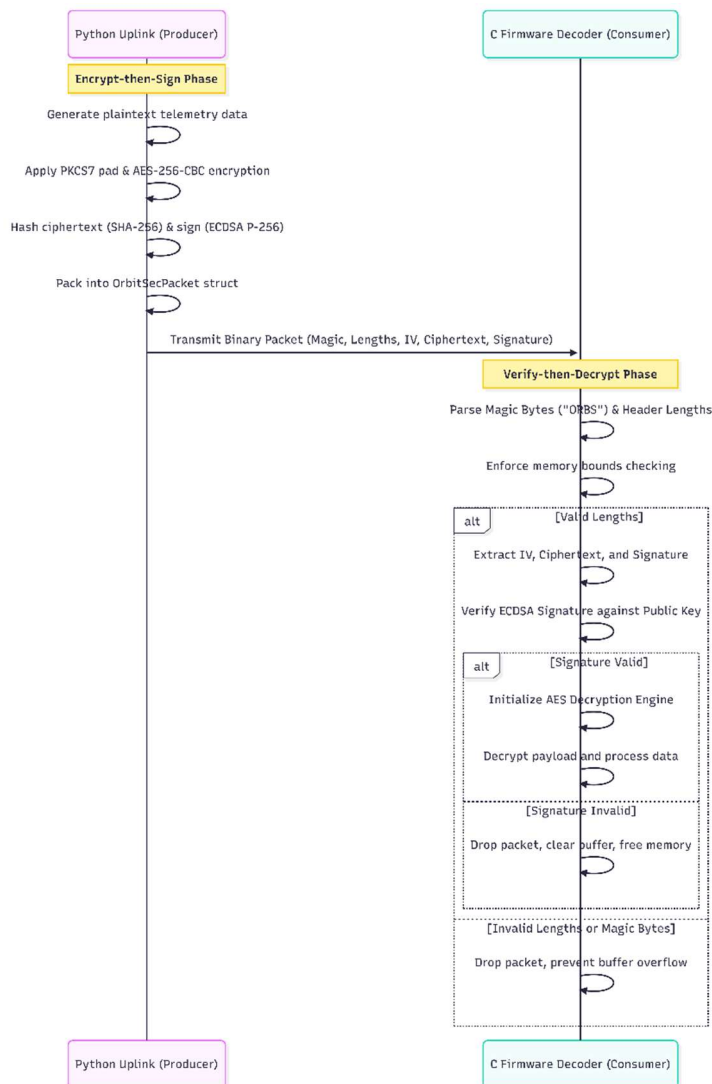
<https://github.com/ctarique/orbitsec>

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1. System Sequence Chart (Swim Lane Diagram)

This diagram illustrates the chronological data flow and state transitions between the transmission source and the receiving device, demonstrating the Encrypt-then-Sign and Verify-then-Decrypt architecture.



2. Functional Requirements

Based on the sequence chart, the following functional constraints define the system's operational parameters.

Requirement ID	Abstract	Notes
R-ORB-Func-1	The Python uplink must encrypt telemetry data using AES-256-CBC.	Data must be padded to exact 16-byte blocks using PKCS7 prior to encryption.
R-ORB-Func-2	The Python uplink must generate an ECDSA signature of the AES ciphertext using the NIST P-256 curve.	Implements the Encrypt-then-Sign paradigm.
R-ORB-Func-3	The system must package data into a strict binary format.	The C struct includes a 4-byte magic identifier, payload length, signature length, 16-byte IV, ciphertext, and signature.
R-ORB-Func-4	The C firmware must parse incoming binary packets as a Finite State Machine.	Simulates receiving chunked byte streams over unstable serial connections.
R-ORB-Func-5	The C firmware must execute the OpenSSL EVP API to decode and decrypt the payload.	Proceeds to decryption only after the security gate passes.

3. Security Requirements

The following requirements map the system's defensive mechanisms to the core OWASP vulnerabilities outlined in the project parameters.

Requirement ID	Abstract	Notes
R-ORB-Sec-1	The firmware must enforce strict bounds checking on all incoming packet lengths.	OWASP: Input Validation / Memory Management. Drops packets exceeding maximum buffer sizes to prevent buffer overflows.
R-ORB-Sec-2	The firmware must validate the ECDSA signature against the public key prior to decryption.	OWASP: Cryptographic Practices / Communication Security. Implements Verify-then-Decrypt to mitigate spoofing and padding oracle attacks.
R-ORB-Sec-3	The system must mandate SHA-256 hashing during the ECDSA signature verification.	OWASP: Data Protection. Ensures bit-level integrity and detects tampering before processing.
R-ORB-Sec-4	The firmware must immediately drop invalid packets and neutralize the AES engine.	OWASP: Error Handling. Occurs upon signature mismatch or invalid magic bytes.
R-ORB-Sec-5	The firmware must explicitly free all dynamically allocated memory and contexts upon failure or completion.	OWASP: Memory Management. Mitigates Denial of Service attacks caused by memory exhaustion.